

REDOX REACTIONS

24/7/24

LN-22

Redox Reaction:

Many chemical reaction involve transfer of electrons from one chemical substance to another. These electrons transfer reactions are termed as oxidation reaction or Redox reaction. OR

Those reactions which involve oxidation and reduction both simultaneously are known as oxidation - reduction or Redox Reaction.

OR

Those reaction in which increase and decrease of oxidation number of same or different atoms occurs are known as Redox Reaction.

Oxidation Number:

Oxidation State of an atom in a molecule or ion is the hypothetical or the real charge present on an atom due to electronegativity difference.

OR

Oxidation state of an element in a compound represents the number of electrons lost or gained during its change from free state into that compound.

Some important points about Oxidation Number:

1. Electro-negativity values of no two elements are same.

eg- $P > H$, $C > H$, $S > C$, $Cl > N$

2. Oxidation number of an element maybe positive or negative.

3. Oxidation number can be 0, whole number or fractional value.

eg. PH_3

$ox.N = -3$

PCl_3

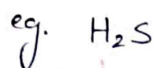
$ox.N = +3$

N_3H

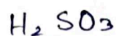
$ox.N = 2 \times 3 + 1 = 0$

$x = -\frac{1}{3}$

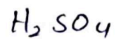
4. Oxidation number of same element can be different in same or different compounds.



oxidation state = -2



O.N = +4



O.N = +6

5. To find the oxidation state, we have to find the central atom first.

choice of central atom:

a. Generally H and O are not considered as central atom.

b. Alkali metals and Halogens are not considered generally.

c. The central atom must have minimum size to hold the surrounding atom.

d. Generally, central atom is that element which is present as least number elements.

e. Central atom are generally electropositive in nature.



$$x + 3(-1) = 0$$

$$x = +3$$



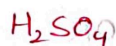
$$x + 2(+1) = 0$$

$$x = -2$$



$$x + 2(-2) = 0$$

$$x = +4$$



$$x + 2(+1) + 4(-2) = 0$$

$$x = +6$$



$$x + 1(+1) + 4(-2) = 0$$

$$x = +7$$



$$x + 1(+1) + 3(-2) = 0$$

$$x = +5$$



$$3 + x - 8 = 0$$

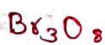
$$x = +5$$

Some Special examples:



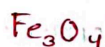
$$2 + x - 8 = 0$$

$$x = +6$$



$$3x + 8(-2) = 0$$

$$x = \frac{16}{3}$$



$$3x - 8 = 0$$

$$x = \frac{8}{3}$$



$$x - 10 = 0$$

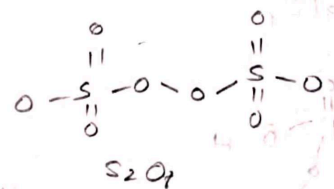
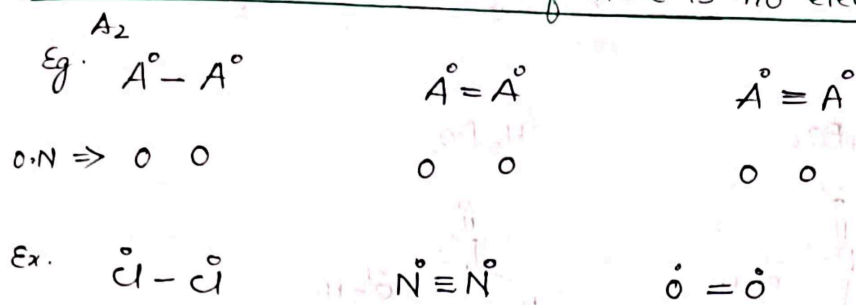
$$x = +10$$

• Rule for Calculating the Oxidation Number:

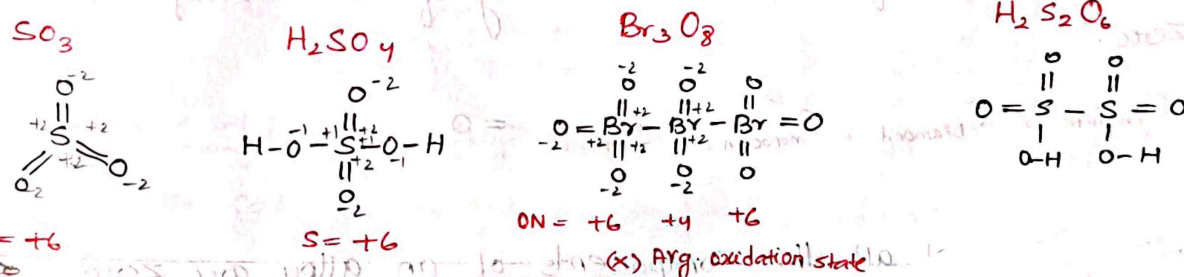
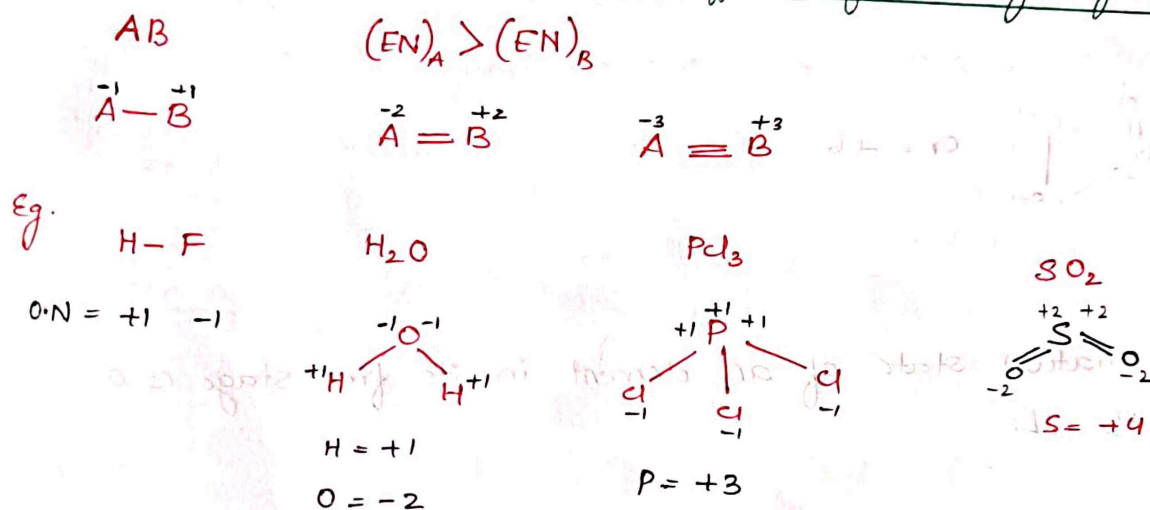
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In case of Covalent bond:

1. For Homoatomic molecule (If there is no electronegativity difference):



2. For Heteroatomic molecule (where difference of electronegativity are there):

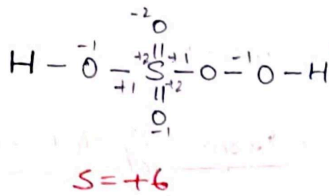


(x) Arg. oxidation state

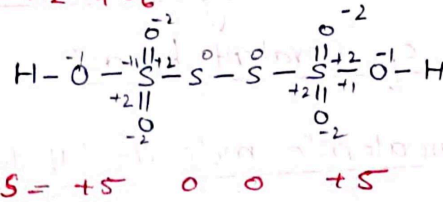
Element	Electronegativity
C	2.5
N	3.0
O	3.5
F	4.0
P	2.2
S	2.5
Cl	3.0
Br	2.8
I	2.5

$F = 4.0$
 $O = 3.5$
 $N = 3.0$
 $Cl = 3.0$
 $Br = 2.8$
 $S = 2.5$
 $I = 2.5$
 $C = 2.5$
 $P = 2.2$

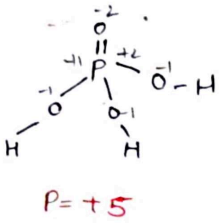
1. H_2SO_4 (Caro's Acid)



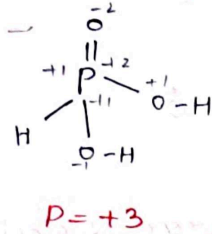
2. $\text{H}_2\text{S}_4\text{O}_6$



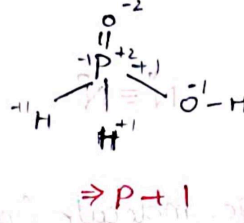
3. H_3PO_4



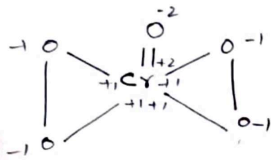
H_3PO_3



H_3PO_2



4. CrO_5



3. The oxidation state of an element in its free stage is 0.
Eg. H, He, Li

4. Oxidation state of an element in any of its allotropic form is zero.

Eg- $\text{C}_{\text{graphite}}$, $\text{C}_{\text{diamond}}$, $\text{S}_{\text{monoclinic}}$, $\text{S}_{\text{rhombic}}$ = 0

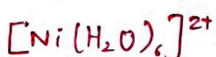
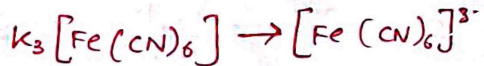
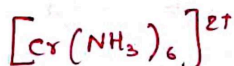
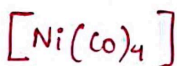
5. Oxidation state of all the components of an alloy are zero.

Eg. $\overset{0}{\text{Na}} - \overset{0}{\text{Hg}}$
amalgam

$\overset{0}{\text{Zn}} - \overset{0}{\text{Hg}}$

6. In complex compounds, oxidation state of some neutral molecules (ligands).

eg. CO, NO, NH_3 , H_2O etc.



$$x + 6(0) = +2$$

$$x + 6(-1) = -3$$

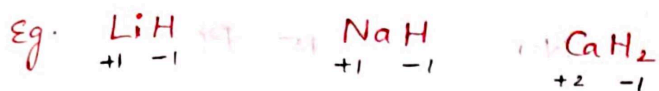
$$x + 6(0) = +2$$

$$x = +2$$

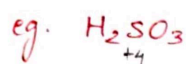
$$x = +3$$

7. Oxidation state of Fluorine in all its compound is -1.

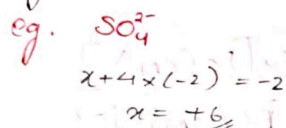
8. Oxidation state of H- in most of its compounds is +1 except in metal hydrides.



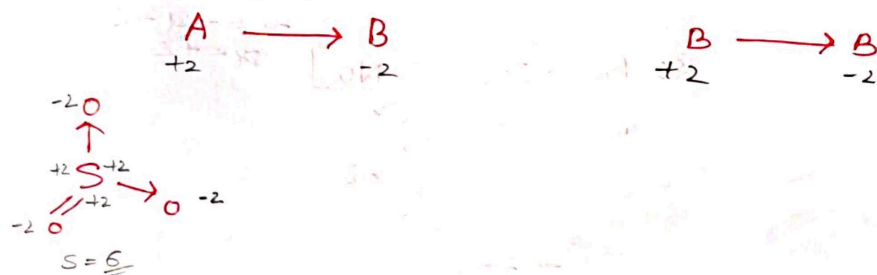
9. The algebraic sum of oxidation state of all the atoms present in a polyatomic neutral molecule is 0.



10. The algebraic sum of oxidation state of all the atoms in a polyatomic ion is equal to the charge present on the ion.



In Case of Co-ordinate bond:



In case of ionic bond:

charge on cation = oxidation state of cation.

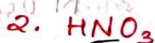
charge on anion = oxidation state of anion.



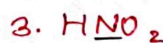
9. Find out the oxidation state of the central atom.



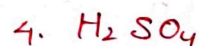
$$x + (2) \times (-2) = 0$$
$$x = +4$$



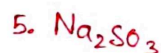
$$1 + x - 6 = 0$$
$$x = +5$$



$$1 + x + 2 \times (-2) = 0$$
$$1 + x - 4 = 0$$
$$x = +3$$



$$2 + x - 8 = 0$$
$$x = +6$$



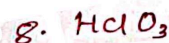
$$(1 \times 2) + x - 6 = 0$$
$$2 + x - 6 = 0$$
$$x = +4$$



$$2 + 1 + x - 8 = 0$$
$$x = +5$$

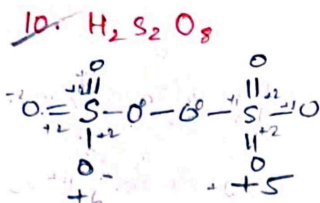


$$1 + x - 8 = 0$$
$$x = +7$$



$$1 + x - 6 = 0$$
$$x = +5$$

9. HClO_2
 $1 + x - 4 = 0$
 $x = +3$



11. $\text{Na}_2\text{S}_4\text{O}_6$
 $\text{S}_4\text{O}_6^{2-}$
 $4x - 12 = -2$

12. $\text{S}_2\text{O}_3^{2-}$

$2x - 6 = -2$
 $2x = -2 + 6$
 $x = +2$

13. N_2O_5

$2x - 10 = 0$
 $x = +5$

14. NO_3^-

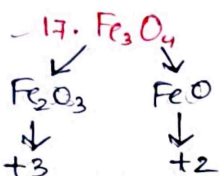
$x - 6 = -1$
 $x = +5$

15. PO_4^{3-}

$x - 8 = -3$
 $x = +5$

16. Fe_2O_3

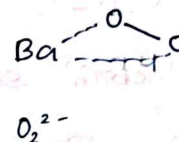
$2x - 6 = 0$
 $x = +3$



18. N_2O_3

$2x - 6 = 0$
 $x = +3$

19. BaO_2



20. Na_2O_2

$2x - 4 = 0$
 $x = +2$

21. NOCl

$x - 2 - 1 = 0$
 $x = +3$

22. $\text{K}_2\text{Cr}_2\text{O}_7$

$2 + 2x - 14 = 0$
 $x = +6$

23. $\text{K}_4[\text{Fe}(\text{CN})_6]$

$[\text{Fe}(\text{CN})_6]^{-4}$
 $x + 6(-1) = -4$
 $x - 6 = -4$
 $x = -4 + 6$
 $x = +2$

24. KIO_3

$1 + x - 6 = 0$
 $x = +5$

25. $\text{H}_2\text{C}_2\text{O}_4$

$2 + 2x - 8 = 0$
 $x = +3$

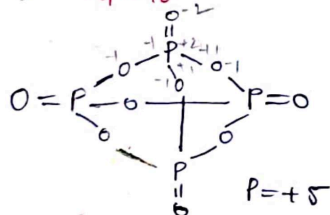
26. $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}]$

$[\text{Fe}(\text{CN})_5\text{NO}]^{-2}$
 $x + 5(-1) + 0 = -2$
 $x - 5 = -2$
 $x = -2 + 5$
 $x = +3$

29. $\text{K}_3[\text{Cu}(\text{CN})_4]$

$[\text{Cu}(\text{CN})_4]^{-3}$
 $x + 4(-1) = -3$
 $x - 4 = -3$
 $x = -3 + 4$
 $x = +1$

27. P_4O_{10}



28. $\text{Ca}_3(\text{PO}_4)_2$

PO_4^{3-}
 $x - 8 = -3$
 $x = +5$

30. $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$

SO_4^{2-} NH_4^+ SO_4^{2-}

$x + 1(-2) + 2(+1) + 1(+2) = 0$
 $x = +2$

31. NodO_4

N

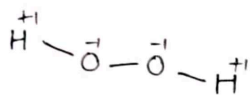
NOTE:

oxidation state of oxygen in most of its compounds is -2 except in —

a. Peroxides (O_2^{2-})

oxidation state = -1

Eg. H_2O_2



BaO_2

O_2^{2-}

b. Superoxides (O_2^{-1})

oxidation state of O = $-\frac{1}{2}$

c. Ozonide (O_3^{-1})

oxidation state of O = $-\frac{1}{3}$

d. O_2F_2

= O_2^{+2}

oxidation state = $+1$

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LN - 24

Hw

1. Oxidation number of cobalt in $[Co(NH_3)_6]Cl_2Br$

$$x + 6(0) - (1 \times 2) - 1 = 0$$

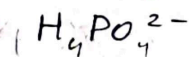
$$x = +3$$

2. The order of increasing oxidation number in S_8 , $S_2O_8^{2-}$, $S_2O_3^{2-}$, $S_4O_6^{2-}$

$S_8 = 0$

$S_2O_8^{2-}$

3. The oxidation number of phosphorus in $Ba(H_2PO_2)_2$



$$= 2x + 4(+1) + 4(-2) = -2$$

$$= 2x + 4 - 8 = -2$$

$$2x = +2$$

$$x = +1$$

APPLICATIONS OF OXIDATION NUMBER:

1. To compare the strength of acid and Base.

Strength of acid $\propto$ oxidation number.

Oxidation no $\propto$ no of valent orbitals.

$\downarrow$
It can accept more e⁻ pair.
 $\downarrow$
acidity $\uparrow$

Q. Compare the acidity of the following compounds.

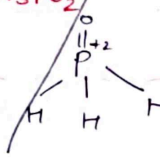
1. HClO_4 HClO_3 HClO_2 HClO

$$\begin{array}{llll} 1+x-8=0 & 1+x-6=0 & 1+x-4=0 & 1+x-2=0 \\ x=+7 & x=+5 & x=+3 & x=+1 \end{array}$$

$\Rightarrow \text{HClO}_4 > \text{HClO}_3 > \text{HClO}_2 > \text{HClO}$

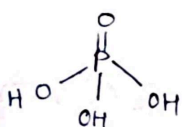
2. H_3PO_4 , H_3PO_3 , H_3PO_2

$$\begin{array}{ll} 3+x-8=0 & 3+x-6=0 \\ x=+5 & x=+3 \end{array}$$

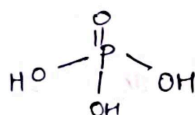


= +1

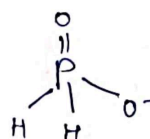
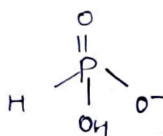
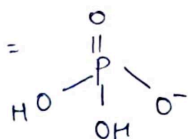
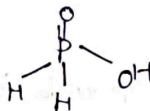
2. H_3PO_4



H_3PO_3



H_3PO_2



* acidity: $\text{H}_3\text{PO}_2 > \text{H}_3\text{PO}_3 > \text{H}_3\text{PO}_4$
increases
with mesomeric.

Strength of base $\propto$ $\frac{1}{\text{oxidation no}}$

2. To determine the oxidising and reducing nature of the substance :

Oxidising agents :

oxidising agents are the substances which accept electrons in a chemical reaction, i.e, electron acceptor are called oxidising agent.

Reducing agents :

These are the substances which donate electrons in a chemical reaction, i.e, electron donors are called reducing agents.

Elements	C	N	P	Cl	Cr	Mn	Os	Ru	O	H	S
Highest Oxidation State	+4	+5	+5	+7	+6	+7	+8	+8	+2	+1	+6
Lowest Oxidation state	-4	-3	-3	-1	0	0	0	0	-2	-1	-2

we can select oxidising or reducing agent by the following rule:

1. If the effective element in a compound is present in maximum oxidation state, then the compound acts as oxidising agent.

Eg. KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$, H_2SO_4 , SO_3 , S_3PO_4 , HNO_3 , HClO_4

2. If effective element in a compound is present in minimum oxidation state, then the compound acts as reducing agent.

Eg- PH_3 , NH_3 , CH_4 , H_2S

3. If effective element in a compound is present in intermediate oxidation state, then the compound can acts as oxidising as well as reducing agent.

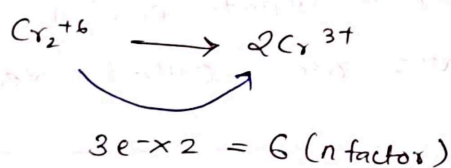
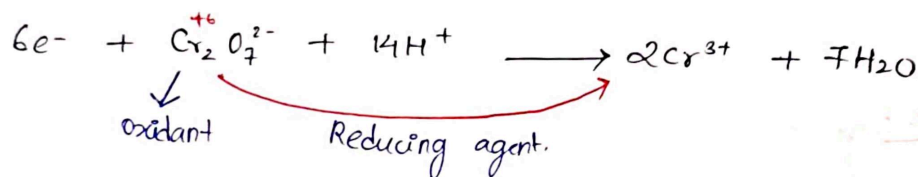
Eg- SO_2 , HNO_2 , H_3PO_3 , H_2O_2

4) To calculate the equivalent weights of an oxidising or reducing agent is that weight which accepts or loses 1 mole of electrons in a chemical reaction.

A. Equivalent weight of oxidant = $\frac{\text{Molecular weight}}{\text{No. of } e^- \text{ gained by 1 mole}}$

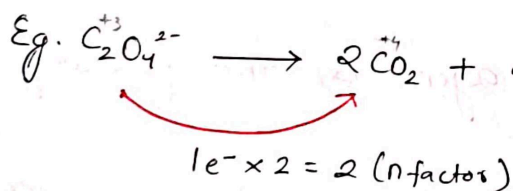
Eg. In acidic medium :

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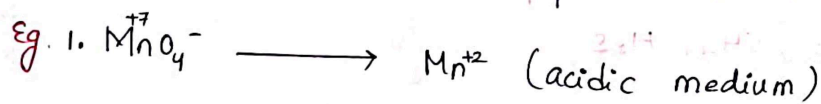
$$\text{Eq. wt} = \frac{M}{6}$$

B. Equivalent weight of reductant = $\frac{\text{Molecular weight}}{\text{No. of } e^- \text{ lost by 1 mole.}}$



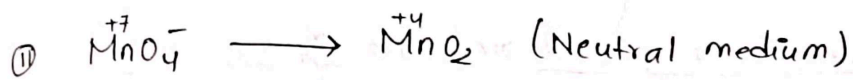
$$\text{Eq. wt} = \frac{M}{2}$$

c. In different conditions, a compound may have different equivalent weight because, it depends upon the number of electrons gained or lost by that compound in that reaction.



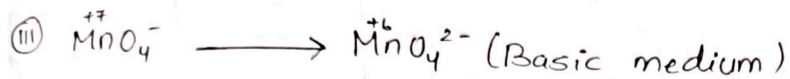
no. of e^- transfer = 5

$$\text{Eq. wt} = \frac{M}{5}$$



no. of e^- transfer = 3

$\text{Eq. wt} = \frac{M}{3}$



no. of e^- transfer = 1

$\text{Eq. wt} = \frac{M}{1}$

Short trick:

B	A	N
1	5	3

NOTE:

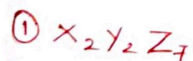
KMnO_4 acts as oxidant in every medium, although with different strength which follows the order-

Acidic medium $>$ Neutral medium $>$ Basic medium

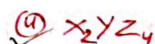
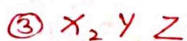
Q. To determine the possible molecular formula of the compound:

Since, the sum of the oxidation number of all the atoms present in a compound is 0, so the validity of the formula can be confirmed.

Q. A compound consist of elements x, y, z . Their oxidation states are $+1, +6$ and -2 respectively. Then what is the molecular formula of that compound.



$2 \times 1 + 2 \times 6 -$



$2 \times 1 + 2 \times 6 - 8 = 0$

$2 + 6 - 8 = 0$

$8 - 8 = 0$

⑤ Both 1 and 4

Some common examples of oxidising and reducing agents with their equivalent weight:

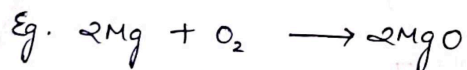
Species	changed To	Reaction	no. of e ⁻ transfer	Eq. wt
MnO_4^- (O.A)	Mn^{2+}	$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$	5	$\frac{M}{5}$
MnO_4^- (O.A)	MnO_2	$\text{MnO}_4^- + 3\text{e}^- + 4\text{H}_2\text{O} \rightarrow \text{MnO}_2 + 4\text{OH}^-$	3	$\frac{M}{3}$
MnO_4^- (O.A)	MnO_4^{2-}	$\text{MnO}_4^- + \text{e}^- \rightarrow \text{MnO}_4^{2-}$	1	$\frac{M}{1}$
$\text{Cr}_2\text{O}_7^{2-}$ (O.A)	Cr^{3+}	$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	6	$\frac{M}{6}$
$\text{S}_2\text{O}_3^{2-}$ (R.A)	$\text{S}_4\text{O}_6^{2-}$	$2\text{S}_2\text{O}_3^{2-} \rightarrow \text{S}_4\text{O}_6^{2-} + 2\text{e}^-$	2 (for 2 moles)	$\frac{2M}{2} = M$
H_2O_2 (O.A)	H_2O	$\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow 2\text{H}_2\text{O}$	2	$\frac{M}{2}$
H_2O_2 (R.A)	O_2	$\text{H}_2\text{O}_2 \rightarrow \text{O}_2 + 2\text{H}^+ + 2\text{e}^-$	2	$\frac{M}{2}$
I^- (R.A)	IO_3^-	$\text{I}^- + 6\text{OH}^- \rightarrow \text{IO}_3^- + 3\text{H}_2\text{O} + 6\text{e}^-$	6	$\frac{M}{6}$

OXIDATION AND REDUCTION:

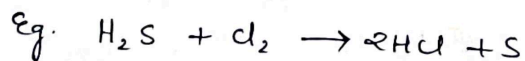
31/7/24
LN-26

OXIDATION (OLD CONCEPT)

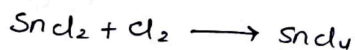
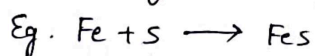
1. Addition of O_2



2. Removal of H



3. Addition of electronegative element



4. Removal of electropositive element



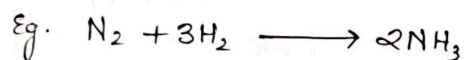
NEW CONCEPT

oxidation process are those process in which one or more e^- are lost by an atom, ion or molecule



REDUCTION (OLD CONCEPT)

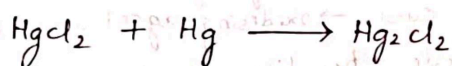
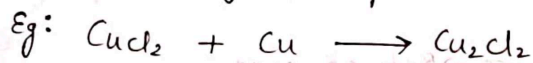
1. Addition of Hydrogen (H_2)



2. Removal of O_2



3. Addition of electropositive element

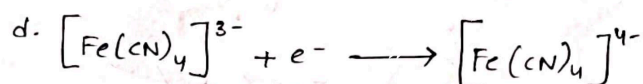
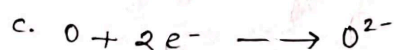
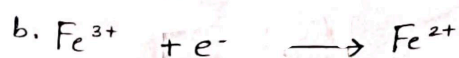
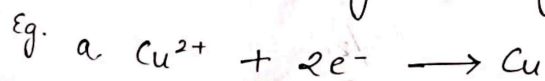


4. Removal of electronegative element.



NEW CONCEPT

Reduction process are those process in which one or more electrons are gained by an atom, ion or molecule.



Redox Reaction

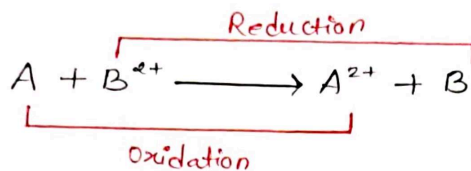
Redox

Oxidation

- addition of O_2
- removal of e^-

Reduction

- addition of H_2
- addition of e^-



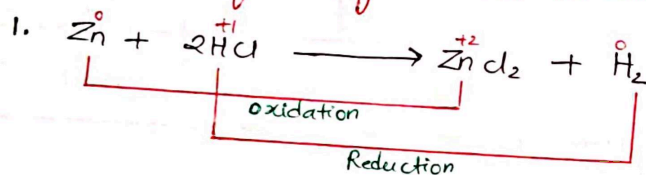
oxidation and
Reduction

↓
Redox Reaction

$A \rightarrow$ Reducing Agent

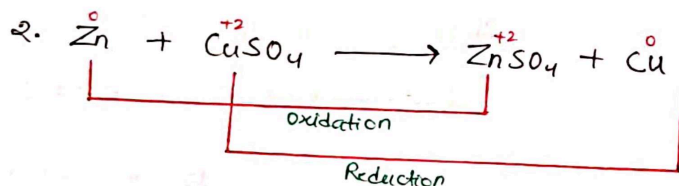
$B^{2+} \rightarrow$ Oxidising Agent

Q. which of the following is not redox reaction.



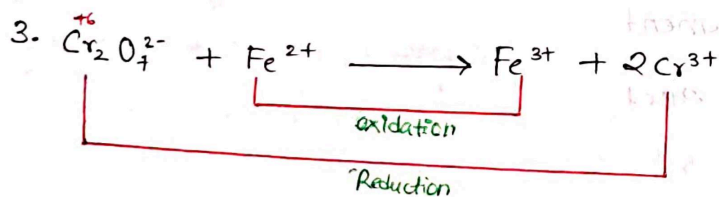
$Zn \rightarrow$ Reducing agent
 $2HCl \rightarrow$ oxidising agent

Redox Reaction



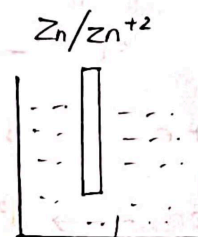
$Zn \rightarrow$ Reducing agent
 $CuSO_4 \rightarrow$ oxidising agent

Redox Reaction



Redox Reaction.

$Fe^{2+} \rightarrow$ Reducing agent
 $Cr_2O_7^{2-} \rightarrow$ oxidising agent



Solution

Zn^{2+}

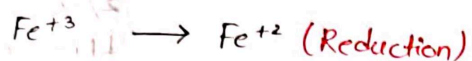
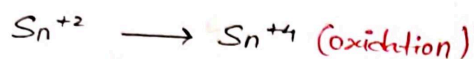
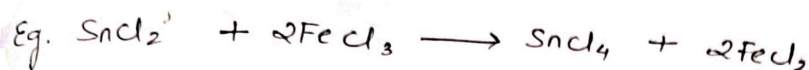
ions $\Rightarrow Zn^{2+}/Ag^+/Cu^{2+}$

Rod $\rightarrow$ Solid Zn or Cu or Ag

TYPES OF REDOX REACTION

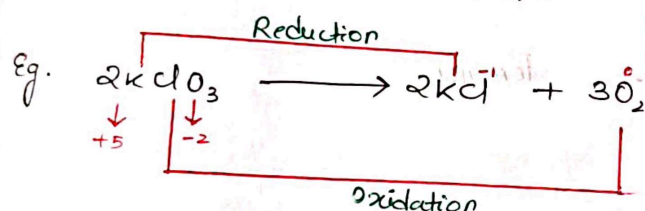
1. INTERMOLECULAR REDOX REACTION

When oxidation and reduction takes place separately in different compounds, then the reaction is called intermolecular redox reaction.



2. INTRA MOLECULAR REDOX REACTION

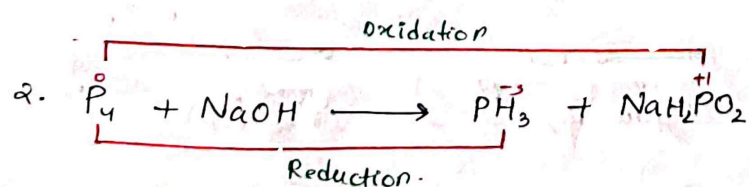
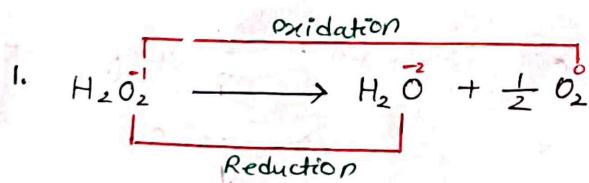
During the chemical reaction, if oxidation and reduction takes place in single compound, then the reaction is called intra-molecular redox reaction.



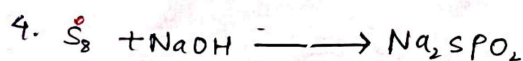
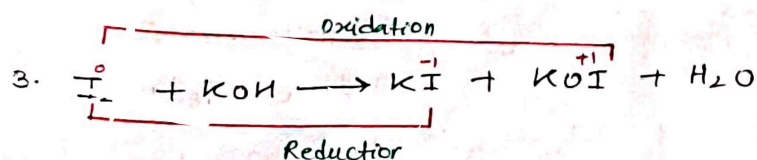
(SAME COMPOUND NOT ELEMENT)

3. DISPROPORTIONATION REACTION

When reduction and oxidation takes place in the same element of the same compound, then the reaction is called disproportionation reaction.

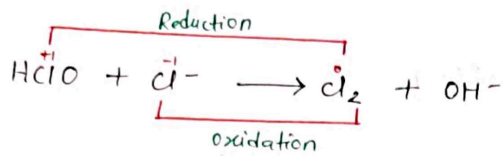
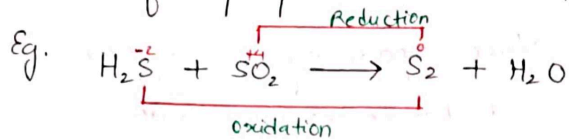


$$\begin{aligned} \text{Na}^+ \text{HPO}_2^- \\ 1 + x + 2(+1) + 2(-2) &= -1 \\ x - 2 &= -1 \\ x &= +1 \end{aligned}$$



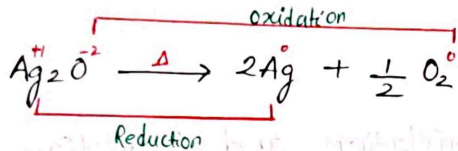
4. COMPROPORTIONATION REACTION

Reverse of disproportionation reaction.

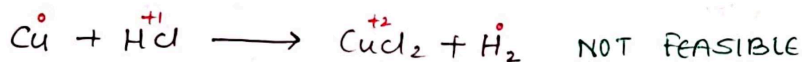
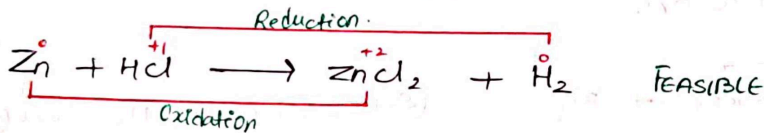


5. DECOMPOSITION REACTION Thermal decomposition.

2/2/24
LN-2



6. METAL DISPLACEMENT REACTION



ELECTROCHEMICAL SERIES

Li	 Oxidation Potential ↑
K	
Ba	
Sr	
Ca	
Na	
Mg	
Al	
Mn	
Zn (-0.76)	
Cr	
Fe	
Co	
Ni	
H → 0	
Cu (+0.34)	 Reduction Potential ↓
Ag	
Au	
I ₂	
Br ₂	
Cl ₂	
F ₂	

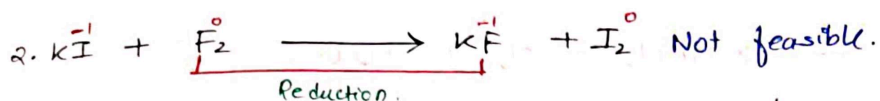
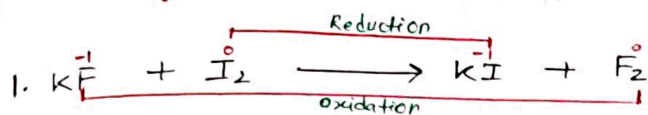
Oxidation Potential ↑
↓
Itself oxidise
↓
Reduce others
↓
Reducing agent
↓
Strongest Reducing agent (Li)

Reduction Potential ↑
↓
Reduce itself
↓
Oxidise others
↓
Oxidising agent
↓
Strongest oxidising agent (F₂)

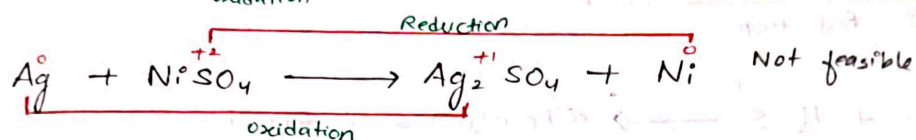
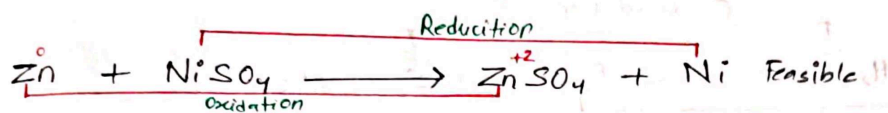
Upper - oxidise

Lower - Reduce

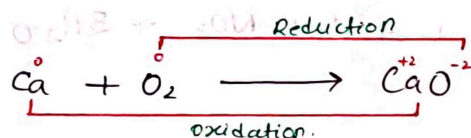
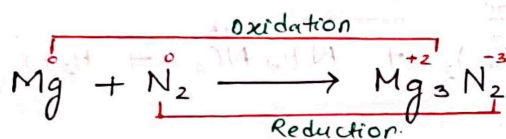
Q. which of the reaction is not feasible?



Q. Can we stir NiSO_4 solution by - ① Zn spoon ② Ag spoon.



7. COMBINATION REACTION OR SYNTHESIS REDOX REACTION :

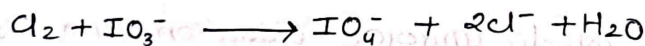
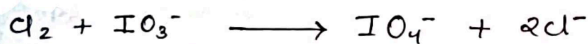
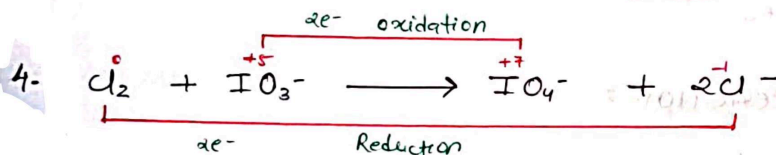
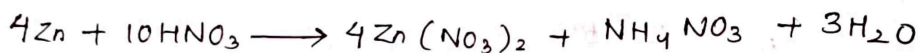
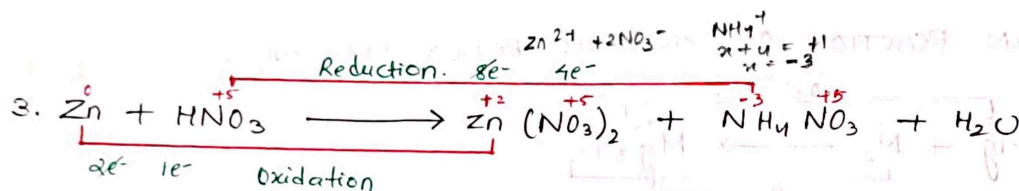
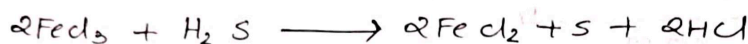
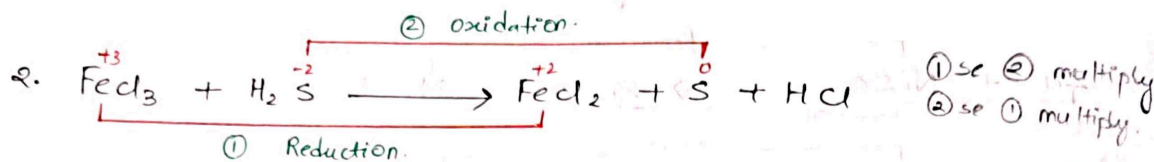
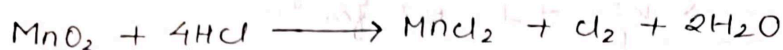
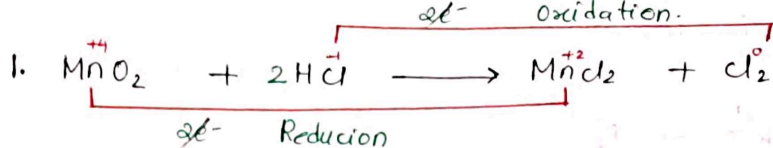


BALANCING OF REDOX REACTION :

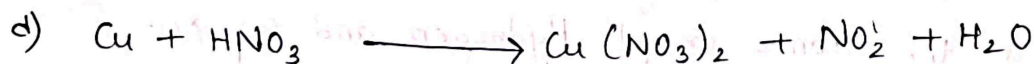
A. OXIDATION STATE METHOD :

Rules:

1. Identify the elements or atoms which undergo oxidation and Reduction.
2. Balance number of atoms which undergo oxidation and reduction.
3. Balance charge by cross multiplication.
4. Balance all the atoms except Hydrogen and oxygen.
5. To balance oxygen and water to the opposite side.
6. To balance Hydrogen add H^+ to the opposite side. (In case - of Acidic Medium)
7. In case of basic medium, we will add OH^- to both - side as many as H^+ ion is present



3/8/24



Rules:

1. Write the equation given to balance possibly in ionic form.
2. Identify elements undergoing oxidation and reduction.

• Oxidation $\rightarrow$ increase in oxidation number

• Reduction $\rightarrow$ decrease in oxidation number.

3. Break the equation into two halves.

A. Oxidation Half

B. Redox Half

4. Balance the half equations by the following ways -

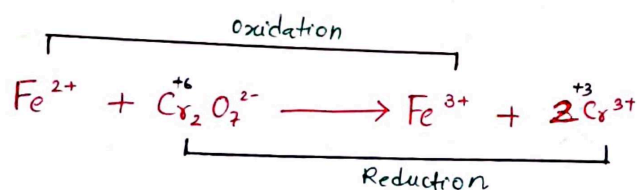
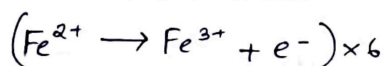
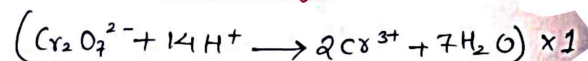
A. Balance all other atoms except oxygen and Hydrogen.

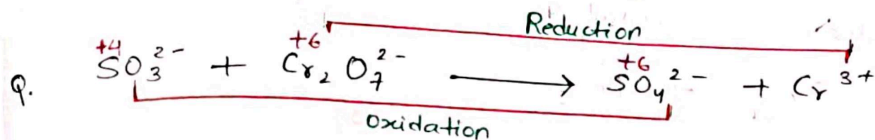
B. Balance oxygen by adding water to the side deficient in oxygen.

C. Balance Hydrogen by H^+ ions.

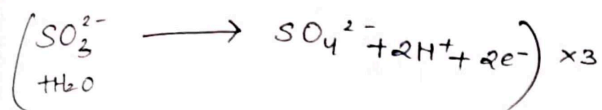
D. Balance charge by electrons.

E. Add the two half such that electrons in two half gets cancelled.

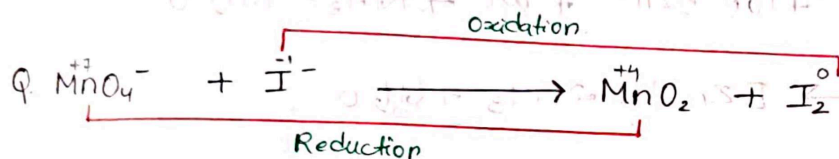
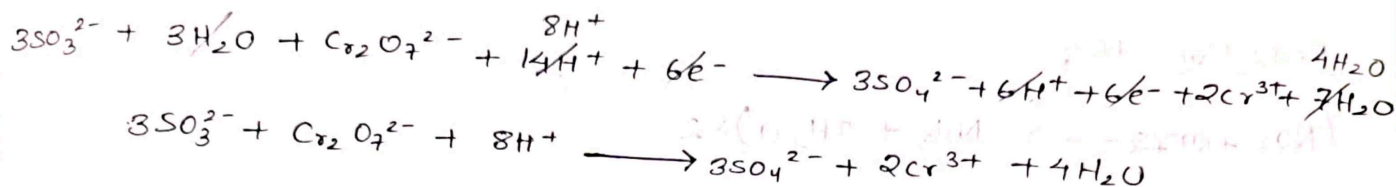
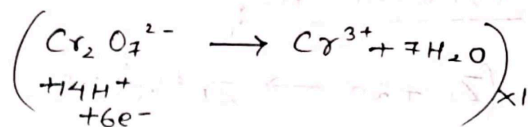
Oxidation Half:Reduction Half



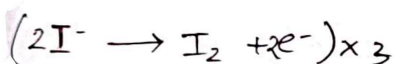
Oxidation Half



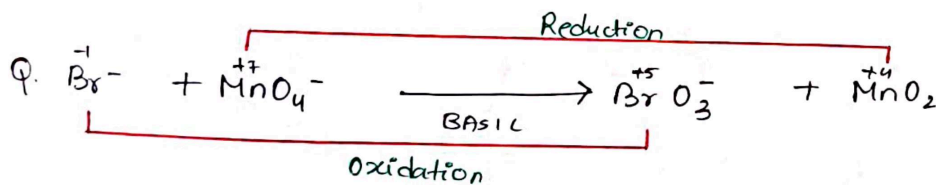
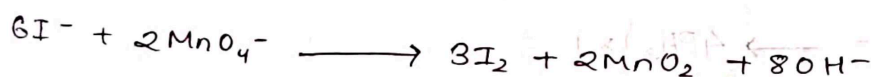
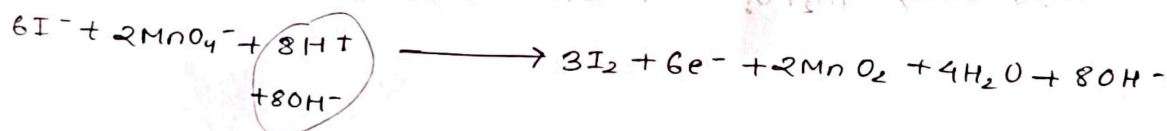
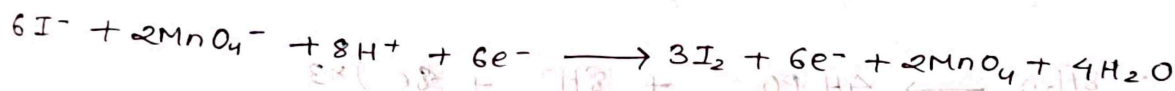
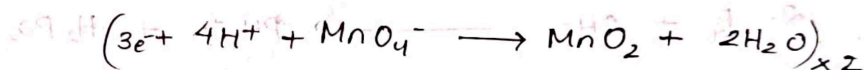
Reduction Half



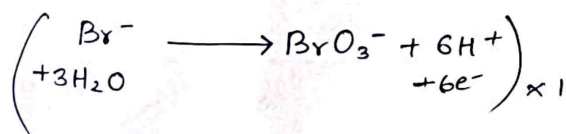
Oxidation Half:



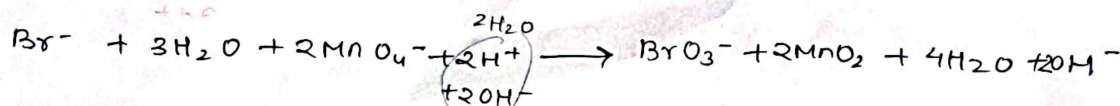
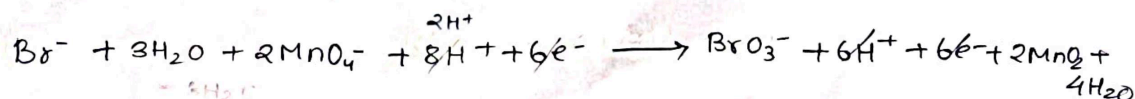
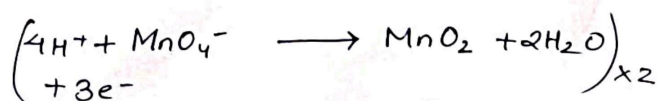
Reduction Half:

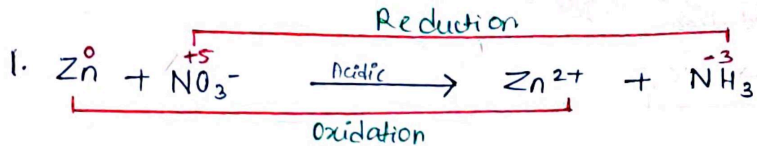


Oxidation Half:



Reduction Half:





Oxidation Half:



Reduction Half:

